

Project 1: Network Reconnaissance and Enumeration Using Nmap

Introduction

This project demonstrates authorized network reconnaissance and enumeration in a controlled home lab using Kali Linux and a Windows virtual machine. The assessment identifies active hosts, open ports, running services and operating system information.

Objectives

Perform reconnaissance and enumeration on an authorized target machine to identify network information, active hosts, open ports, running services, and operating system details.

Scope

Testing was performed only on systems owned by the tester within a private home lab.

Lab Environment

COMPONENTS	DETAILS
Attacker	Kali Linux (10.124.240.71)
Target	Windows VM (10.124.240.49)
Network	Network Bridge Adaptor
Hypervisor	VirtualBox

Methodology

IP identification, gateway verification, host discovery, connectivity testing, TCP port scanning, service version detection, default script scanning, operating system detection, comprehensive scan, and analysis. Practical Implementation.

1. Identify Kali IP

Command:

`ip addr`

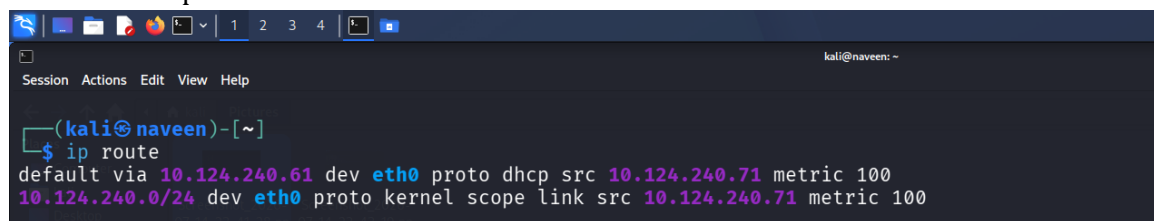
```
(kali@naveen)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 10.124.240.71/24 brd 10.124.240.255 scope global dynamic noprefixroute eth0
        valid_lft 3496sec preferred_lft 3496sec
    inet6 2402:8100:2869:9c1f:895e:b6de:143e:e63a/64 scope global dynamic noprefixroute
        valid_lft 7099sec preferred_lft 7099sec
    inet6 fe80::e709:84b4:fca1:5bec/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Comments:

The kali IP was successfully identified using the `ip addr` command , I have found my IP address 10.124.240.49

2. Verify Gateway

Command: `ip route`



```
(kali@naveen)-[~]
$ ip route
default via 10.124.240.61 dev eth0 proto dhcp src 10.124.240.71 metric 100
10.124.240.0/24 dev eth0 proto kernel scope link src 10.124.240.71 metric 100
```

Comments:

The routing table and default gateway were verified successfully.

3. Identify Configuration file

Command: `cat /etc/resolv.conf`

```
(kali@naveen)-[~]
$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 10.124.240.61
nameserver 2402:8100:2869:9c1f::40
```

Comments:

DNS configuration was verified using `cat /etc/resolv.conf` command. The configured DNS servers were successfully identified.

4. Connectivity Test

Command: ping google.com

```
(kali@naveen)-[~]
$ ping google.com
PING google.com (2404:6800:4000:100e::8b) 56 data bytes
64 bytes from pu-in-f139.1e100.net (2404:6800:4000:100e::8b): icmp_seq=1 ttl=108 time=57.2 ms
64 bytes from pu-in-f139.1e100.net (2404:6800:4000:100e::8b): icmp_seq=2 ttl=108 time=84.9 ms
64 bytes from pu-in-f139.1e100.net (2404:6800:4000:100e::8b): icmp_seq=3 ttl=108 time=84.7 ms
64 bytes from pu-in-f139.1e100.net (2404:6800:4000:100e::8b): icmp_seq=4 ttl=108 time=63.0 ms
64 bytes from pu-in-f139.1e100.net (2404:6800:4000:100e::8b): icmp_seq=5 ttl=108 time=91.8 ms
^C
--- google.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4005ms
rtt min/avg/max/mdev = 57.156/76.316/91.822/13.633 ms
```

Comments:

Internet connectivity and DNS resolution were successfully verified.

5. Host Discovery

Command: nmap -sn 10.124.240.0/24

```
kali@naveen: ~
Session Actions Edit View Help
(kali@naveen)-[~]
$ nmap 10.124.240.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:14 -0400
Nmap scan report for 10.124.240.49
Host is up (0.016s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)

Nmap scan report for 10.124.240.61
Host is up (0.0037s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: C2:DC:A9:3E:4C:ED (Unknown)

Nmap scan report for 10.124.240.207
Host is up (0.00052s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
7070/tcp  open  realserver
MAC Address: 04:EA:56:38:75:DE (Intel Corporate)

Nmap scan report for 10.124.240.71
Host is up (0.0030s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 256 IP addresses (4 hosts up) scanned in 19.54 seconds
```

Comments:

The host discovery scan successfully identified all active devices within the specified network range. This confirmed network connectivity and provided the list of reachable hosts for further enumeration.

6. TCP Port Scan

Command: nmap 10.124.240.49

```
(kali@naveen)-[~]
$ nmap 10.124.240.49
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:44 -0400
Nmap scan report for 10.124.240.49
Host is up (0.0033s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 5.46 seconds
```

Comments:

A TCP Port scan was performed using nmap. The scan was successfully identified the accessible

7. Service Version Detection

Command: nmap -sV 10.124.240.49

```
(kali@naveen)-[~]
$ nmap -sV 10.124.240.49
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:18 -0400
Nmap scan report for 10.124.240.49
Host is up (0.0037s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
445/tcp    open  microsoft-ds?
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.76 seconds
```

Comments:

Service version running on the detected open ports were successfully identified.

8. Operating System Detection

Command: sudo nmap -O 10.124.240.49

```
(kali@naveen)-[~]
$ sudo nmap -O 10.124.240.49
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:20 -0400
Nmap scan report for 10.124.240.49
Host is up (0.0013s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 10|11|2019 (92%)
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2019
Aggressive OS guesses: Microsoft Windows 10 1903 - 22H2 (92%), Microsoft Windows 10 22H2 (92%), Microsoft Windows 10 1803 (91%), Microsoft Windows 11 (87%),
Microsoft Windows 10 1909 (85%), Microsoft Windows 10 1909 - 2004 (85%), Microsoft Windows Server 2019 (85%), Microsoft Windows 10 1809 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

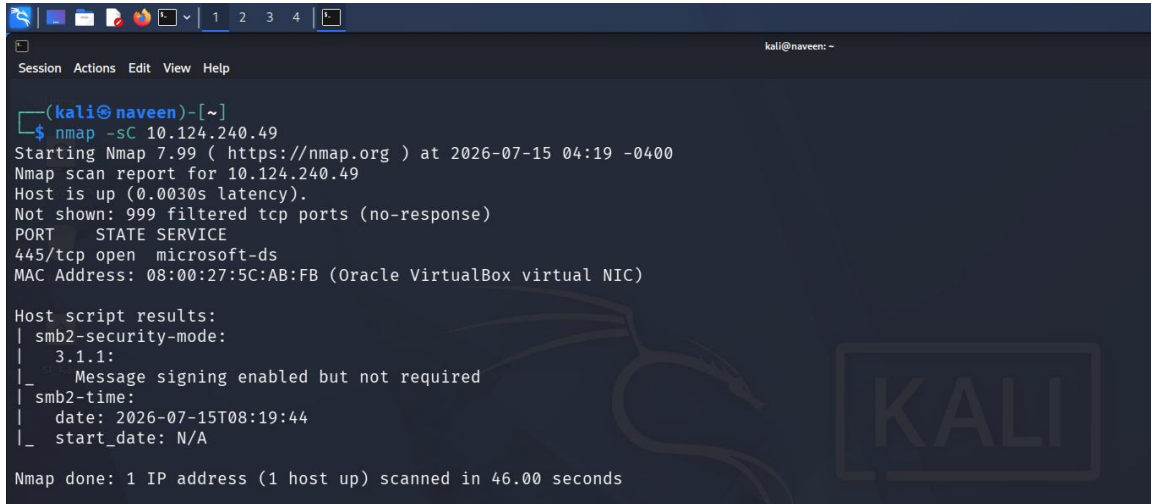
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.51 seconds
```

Comments:

Operating system detection was performed to estimate the target OS

9. Default Script Scan

Command: nmap -sC 10.124.240.49



```
(kali@naveen)-[~]
$ nmap -sC 10.124.240.49
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:19 -0400
Nmap scan report for 10.124.240.49
Host is up (0.0030s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp    open  microsoft-ds
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
| smb2-time:
|   date: 2026-07-15T08:19:44
|_  start_date: N/A

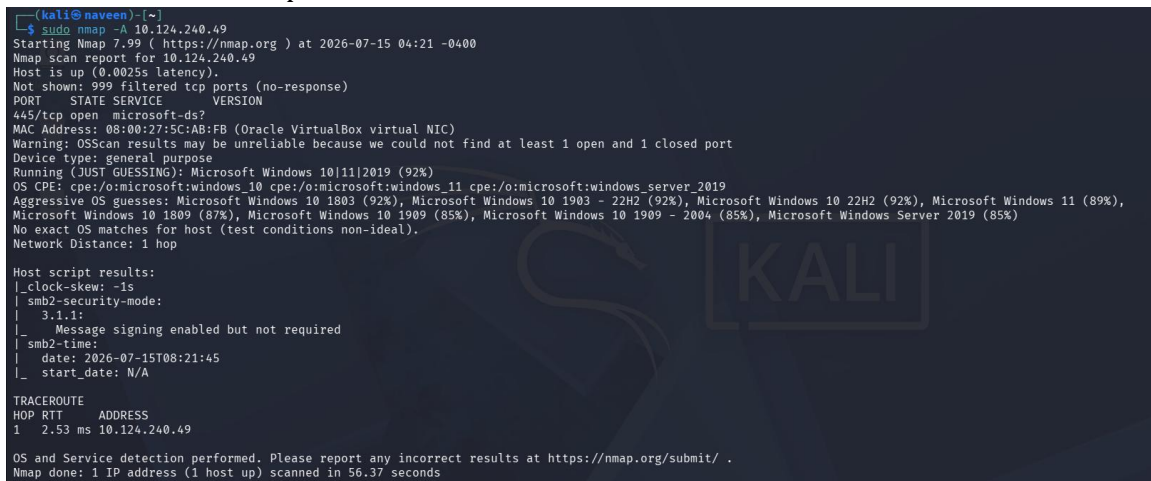
Nmap done: 1 IP address (1 host up) scanned in 46.00 seconds
```

Comments:

Default NSE scripts were executed to gather additional information about the target.

10. Comprehensive Scan

Command: sudo nmap -A 10.124.240.49



```
(kali@naveen)-[~]
$ sudo nmap -A 10.124.240.49
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:21 -0400
Nmap scan report for 10.124.240.49
Host is up (0.0025s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
445/tcp    open  microsoft-ds?
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 10[11|2019] (92%)
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2019
Aggressive OS guesses: Microsoft Windows 10 1803 (92%), Microsoft Windows 10 1903 - 22H2 (92%), Microsoft Windows 10 22H2 (92%), Microsoft Windows 11 (89%),
Microsoft Windows 10 1809 (87%), Microsoft Windows 10 1909 (85%), Microsoft Windows 10 1909 - 2004 (85%), Microsoft Windows Server 2019 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

Host script results:
|_ clock-skew: -1s
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required
| smb2-time:
|   date: 2026-07-15T08:21:45
|_  start_date: N/A

TRACEROUTE
HOP RTT      ADDRESS
1   2.53 ms  10.124.240.49

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 56.37 seconds
```

Comments:

An Advanced scan was performed, including OS detection, Service version detection, default Scripts, and traceroute.

11. Save Scan Report

Command: sudo nmap -A -oN Project01_Scan.txt 10.124.240.49

```

kali@kali:~$ sudo nmap -A -oN Project01_Scan.txt 10.124.240.49
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-15 04:22 -0400
Nmap scan report for 10.124.240.49
Host is up (0.0012s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT      STATE SERVICE
445/tcp    open  microsoft-ds?
MAC Address: 08:00:27:5C:AB:FB (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running (JUST GUESSING): Microsoft Windows 10(11)(2019 (92%))
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2019
Aggressive OS guesses: Microsoft Windows 10 1803 (92%), Microsoft Windows 10 1903 - 22H2 (92%), Microsoft Windows 10 22H2 (92%), Microsoft Windows 11 (89%), Microsoft Windows 10 1909 (85%),
Microsoft Windows 10 1909 - 2004 (85%), Microsoft Windows Server 2019 (85%), Microsoft Windows 10 1809 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

Host script results:
|_clock-skew: -1s
|_smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
|_smb2-time:
|   date: 2026-07-15T08:23:13
|   start_date: N/A

TRACEROUTE
Hop RTT ADDRESS
1 1.20 ms 10.124.240.49

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 56.46 seconds

```

Comments:
The advanced scan results were successfully saved to day1.txt for future analysis and documentation.

Results
The assessment successfully identified the target host, active network services, open ports, and operating system information. The collected data forms the basis for further security assessment.

Target Host IP	10.124.240.49
Open Ports	01
Running Services	Microsoft-ds
Operating System	VirtualBox virtual NIC

Recommendations
Disable unnecessary services, keep systems updated, restrict unused ports using firewall rules, and perform periodic security assessments.

Conclusion
The project successfully demonstrated network reconnaissance and enumeration against an authorized home lab. The assessment identified network information, active hosts, open ports, running services, and operating system details while following ethical testing practices.